

Keuntae Kim

PhD Candidate, Department of Computer Science, Hanyang University

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Research Summary

PhD Candidate at Hanyang University researching how large language and multimodal models reason, adapt continually, and remain grounded in evidence. My primary focus is reasoning in LLMs and VLMs—including reinforcement learning and policy optimization, continual / incremental learning, and retrieval-augmented and meta-prompted reasoning. I extend this line of work to diffusion language and multimodal models (dLLMs, dMLLMs), Video-LLMs, Vision-Language-Action (VLA) models, and world models. I serve as lead researcher on two Korean national AI projects (NRF, IITP) on generative AI, and as a reviewer for NeurIPS, ICML, CVPR, ECCV, and ACL ARR.

Education

Hanyang University, Ph.D. in Computer Science

Sep. 2023 – Present

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Hanyang University, M.S. in Computer Science

Sep. 2021 – Aug. 2023

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Kookmin University, B.S. in Automobile and IT Convergence

Mar. 2015 – Aug. 2019

Seoul, Korea

Research Interests

- LLMs / VLMs Reasoning
- Diffusion Large/Multimodal Language Models (dLLMs, dMLLMs)
- Video-LLMs, World Models, Vision-Language-Action models
- Continual / Incremental Learning
- Reinforcement Learning & Policy Optimization

Selected Publications

Bold + underline = author; * = first author.

1. **Keuntae Kim***, Mingyu Kang, Yong Suk Choi. **Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models.** *CVPR 2026*. **PUBLISHED**
2. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. **ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding.** *ICML 2026*. **PUBLISHED**
3. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. **CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections.** *NeurIPS 2025*. **PUBLISHED**

Publications (Journals & Conferences)

1. **Keuntae Kim***, Yong Suk Choi. “CFP-AL: Combining Model Features and Prediction for Active Learning in Sentence Classification.” *Applied Sciences*, 2025. **PUBLISHED**
2. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections.” *NeurIPS 2025*. **PUBLISHED**
3. **Keuntae Kim***, Mingyu Kang, Yong Suk Choi. “Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models.” *CVPR 2026*. **PUBLISHED**
4. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding.” *ICML 2026*. **PUBLISHED**
5. **Keuntae Kim***, Eunhye Jung, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “Analyzing Creative Capabilities in RLVR Models: Overconfidence and High-Entropy Segments.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
6. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “Revisiting Incremental Learning: A Three-Interface Diagnosis of Stability and Plasticity.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**

7. **Keuntae Kim**^{*}, Jeonghyun Kim, Dongjun Kim, HyeongSeok Kang, Yong Suk Choi. “Projector-JEPA: A Gated DeltaNet MoE Projector Bridging Video-LLMs to World Understanding.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
8. JungHyoun Kim, **Keuntae Kim**, HyeongSeok Kang, Dongjun Kim, Yong Suk Choi. “FourDelta: Fourier-Band Delta Predictor for Latent World Models.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
9. **Keuntae Kim**^{*}, Eunhye Jeong, Yong Suk Choi. “CWM: Controllable White-Box Meta-Prompting for Adaptive Retrieval-Augmented Generation and Reasoning Ability.” *EMNLP 2026 (under review)*. **UNDER REVIEW**

Patents

1. “Method for Analysing Emotion Based on SNS Data,” KR App. No. 10-2023-0193546.
2. “Method and Device for Displaying Emotion Map Animation,” KR App. No. 10-2024-0152245.
3. “Method and System for Fusion of Sparse-Dense and Global Attention for Video Action Recognition,” KR App. No. 10-2024-0132801.
4. “Device and Method for Correcting Toxic Words in Sentence,” KR App. No. 10-2024-0143175.

Research Projects & Grants

- **Metacognitive AGI Framework and its Applications**, IITP, Korea — *Lead Researcher* (Jul. 2025 – Present).
- **Metacognitive Generative AI Framework for Mitigating Hallucination in LLMs**, NRF, Korea — *Lead Researcher* (Feb. 2025 – Present).
- **Big Data-based Real-time Monitoring and Social Simulation System for Risk Management in a Hyper-connected Society**, NRF, Korea — *Lead Researcher, Group 3* (Sep. 2021 – Feb. 2025).
- **High-performance Deep Learning via Cross-connection of Autoencoders and Decoders**, NRF, Korea — *Researcher* (Sep. 2021 – Feb. 2024).
- **Natural Language Data Refinement for Large-scale LM Construction**, KT Corp., Korea — *Researcher* (Sep. 2021 – Jul. 2022).

Technical Skills

AI & Research. Prompt Engineering, LLMs, VLMs, dLLMs / dMLLMs, Video-LLMs, World Models, RAG, VLA

Training. DeepSpeed, HF Accelerate, PEFT (LoRA), bitsandbytes, TRL, Megatron-LM, ms-swift

Serving. vLLM, TensorRT-LLM, Triton, ONNX Runtime, HF TGI, SGLang, llama.cpp

Frameworks. PyTorch, JAX, TensorFlow, Hugging Face, scikit-learn, LangChain

Languages. Python, C/C++, CUDA (*core*) • Java, Kotlin, JavaScript, Rust, Go, MATLAB, SQL (*other*)

Infra & MLOps. Docker, Kubernetes, Git, Linux, CI/CD (Jenkins, GitHub Actions), MLflow, W&B • AWS, GCP, Azure

Data. MySQL, PostgreSQL, MongoDB, Apache Spark, Hadoop

Awards & Honors

- NeurIPS 2025 Scholar Award
- BK21 FOUR Scholarship (Korean government-funded research fellowship), 2021 – Present

Academic Service

Conference & Journal Reviewer

- **NeurIPS 2026** — Conference on Neural Information Processing Systems (Reviewer)
- **ICML 2026** — International Conference on Machine Learning (Reviewer)
- **CVPR 2026** — IEEE/CVF Conference on Computer Vision and Pattern Recognition (Reviewer)
- **ECCV 2026** — European Conference on Computer Vision (Reviewer)
- **ACL ARR — March 2026** — ACL Rolling Review (Reviewer)
- **ACL ARR — January 2026** — ACL Rolling Review (Reviewer)